

Final report

RUN · 9A8885CB · HOW DOES MISSISSAUGA HANDLE A -30°C COLD SNAP WITH AGGRESSIVE ELECTRIFICATION BY 2050?

SCENARIOS	FEEDERS MODELED	CUSTOMERS TESTED	AGENT RUNTIME 1M 28S
1	176	192,544	

VERDICT (WORST CASE)

FAIL

Network fails. 22 feeder(s) past thermal limit; system peaks at 5117 MW (-4% headroom).

PEAK DEMAND

5117 MW

@ hour 43 · -32.5°C

HEADROOM AT PEAK

-4%

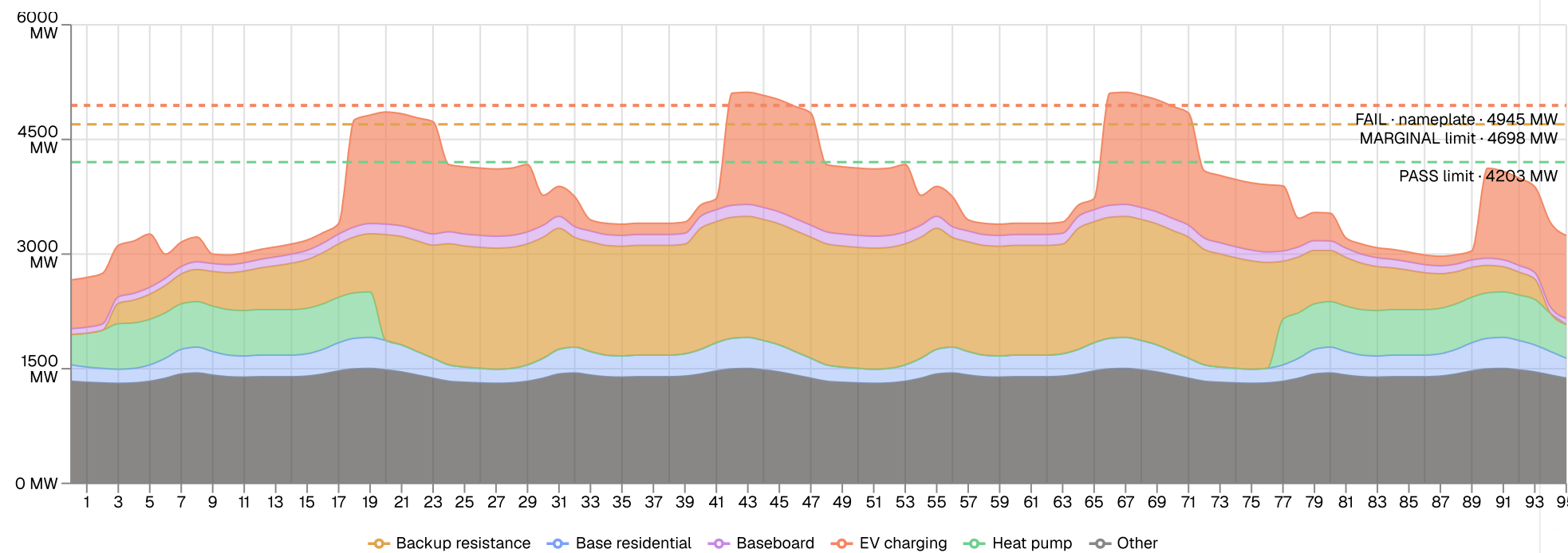
-172 MW remaining

01 · SCENARIO LOAD CURVES

Hourly load profile per scenario, with peak hour and headroom.



Worst-case scenario: Aggressive · 2050 (90% HP, 95% EV) · peak 5117 MW



Aggressive · 2050 (90% HP, 95% EV)

FAIL

PEAK LOAD

5117 MW @ hour 43 (-32.5°C)

HEADROOM

-4% (-172 MW)

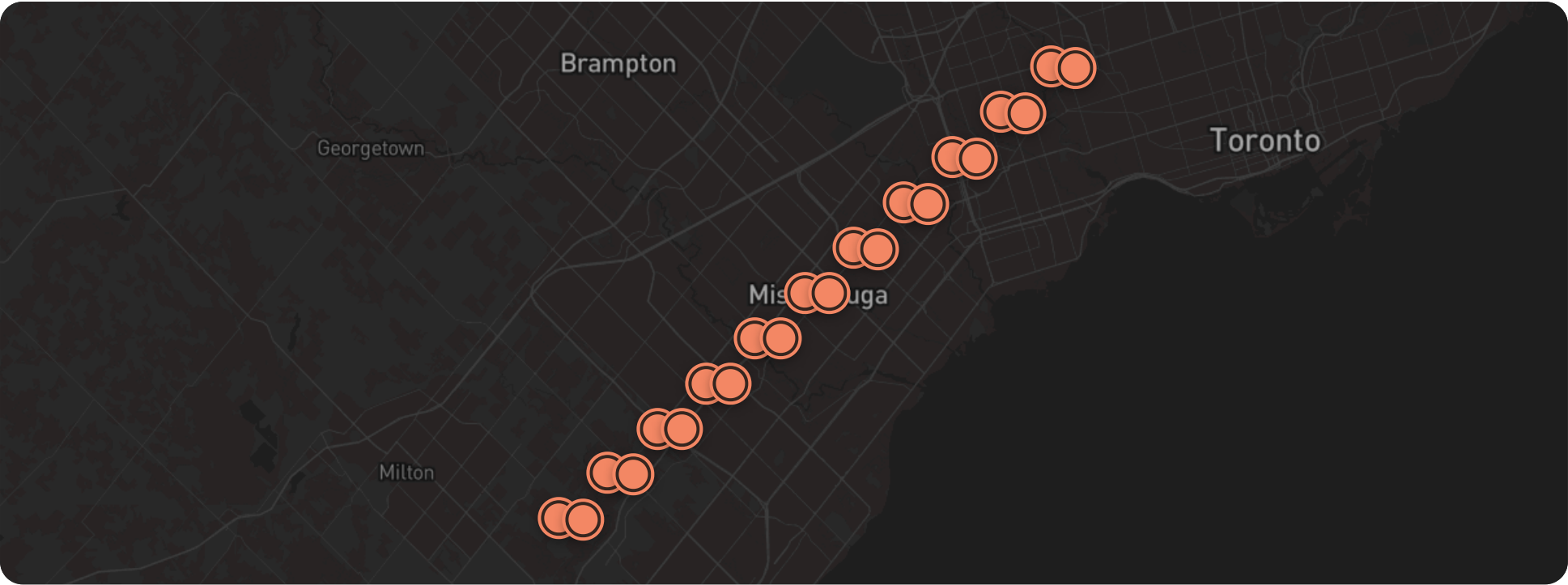
AT-RISK FEEDERS

88 of 176

02 · AT-RISK SUBSTATIONS

Substation risk map with the top feeders ranked by overload risk.

Substation risk map under **Aggressive · 2050 (90% HP, 95% EV)**. Click a marker for capacity + feeder counts.



● CRITICAL ≥ 75% ● AT RISK 50-74% ● STABLE < 50%

Top 10 of 176 feeders, ranked by overload risk.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
1	Mississauga TS 01 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
2	Mississauga TS 02 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).
3	Mississauga TS 03 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).
4	Mississauga TS 04 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).
5	Mississauga TS 05 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).

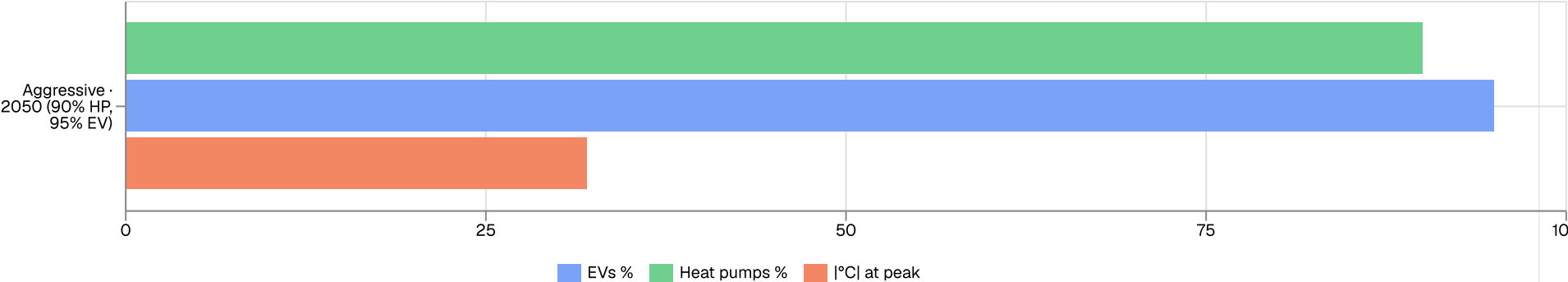
RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
6	Mississauga TS 06 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).
7	Mississauga TS 07 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).
8	Mississauga TS 08 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).
9	Mississauga TS 09 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
10	Mississauga TS 10 F05	117%	93%	thermal overload	This 30-year-old feeder (age_years: 30) with no capacity upgrades (had_upgrade: false) is already operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66), and under the 90% heat-pump, 95% EV scenario its currently low heat-pump share (heat_pump 5%, heating_mix field) and 4.2% EV penetration (ev_pct: 4.2) would surge dramatically, driving thermal overload (failure_mode: thermal_overload) across all 1,364 customers (customers: 1364).

03 · MITIGATIONS & METHODOLOGY

Scenario severity, recommended interventions, methodology notes.

Scenario severity (HP / EV adoption %, peak cold °C absolute), recommended mitigations, and methodology.



RECOMMENDED MITIGATIONS

DEMAND RESPONSE

EV Managed Charging Demand Response Program

Relief: **85.0 MW** Cost: **\$4.5M** 12mo

At 95% EV penetration (ev_pct: 0.95 in the Aggressive scenario) across 192,544 customers (customers: 192544), a managed charging program that defers non-critical EV charging during hours 40 to 50 of a cold event (bracketing the modeled peak at hour 43, peak_hour: 43) can shift a meaningful share of EV load to off-peak hours. Enrolling approximately 15,000 customers at an industry-typical program cost of \$300 per enrolled customer yields an estimated program cost of \$4.5M CAD. Load relief of 85 MW is a conservative estimate based on typical 6 kW average EV charger load and a 50% deferral compliance rate across enrolled customers. This is the fastest deployable mitigation and should be the first action taken.

TRANSFORMER UPGRADE

LOAD SHIFT

Heat Pump Cycling Agreement with Smart Thermostat Partners

Relief: **40.0 MW** Cost: **\$1.8M** 18mo

At 90% heat pump penetration (hp_pct: 0.9 in the Aggressive scenario), coordinated 15-minute cycling of heat pump compressors during the peak window can reduce coincident demand without materially affecting indoor temperatures given typical building thermal mass. Enrolling approximately 12,000 customers via smart thermostat integration at an estimated \$150 per customer for program administration and hardware incentives yields a cost of \$1.8M CAD. A 40 MW relief estimate assumes a conservative 3.3 kW average heat pump draw and 10% of enrolled units cycling at any given moment. This measure complements EV demand response and targets a different load class.

FEEDER RECONDUCTOR

FEEDER RECONDUCTOR

Reconductor and Upgrade Three MIS-F05 Feeders

Relief: **30.0 MW** Cost: **\$4.5M** 30mo

Feeders MIS-01-F05, MIS-02-F05, and MIS-03-F05 are each operating at 117.3% utilization (utilization_pct: 117.3) against a 30.66 MVA rating (rated_mva: 30.66) and are 30 years old (age_years: 30) with no prior upgrades (had_upgrade: false). Reconductoring all three to a higher-rated conductor would restore positive headroom on these circuits and protect the combined 4,092 customers (3 feeders x 1,364 customers each) they serve. At a conservative Canadian reconductor cost of \$1.5M per feeder, the three-feeder program totals \$4.5M CAD. The 30 MW relief figure reflects the incremental capacity headroom restored by upgrading each feeder from its current 30.66 MVA rating to a typical next-tier conductor rating, though the exact upgraded rating should be confirmed through detailed engineering.

Substation Transformer Capacity Additions at Two Highest-Stress Substations

Relief: **60.0 MW** Cost: **\$8.0M** 48mo

With 88 of 176 feeders (at_risk_feeders: 88) flagged as at risk, the upstream substations feeding the highest concentration of at-risk feeders are likely capacity-constrained. Adding transformer capacity at two priority substations (selected from the 22 total, substations: 22, based on feeder concentration analysis to be completed) at an estimated \$4M per substation transformer addition yields a total cost of \$8M CAD. A 60 MW relief estimate is based on industry rule-of-thumb transformer additions of 30 MW per unit in comparable urban Ontario distribution networks. This is a longer-lead capital item and should be initiated in parallel with near-term demand response programs, not instead of them.

Targeted Battery Storage Deployment on At-Risk Feeders

Relief: **25.0 MW** Cost: **\$6.0M** 24mo

Deploying grid-scale battery storage (approximately 5 MW per unit, 4-hour duration) at five of the 88 at-risk feeder endpoints would provide 25 MW of dispatchable peak shaving capacity during the critical hours 40 to 50 window identified by the model (peak_hour: 43). At an estimated \$1.2M CAD per 5 MW, 4-hour system installed cost (consistent with recent Canadian utility procurement ranges), five units total \$6M CAD. This option also improves resilience for the 96-hour event duration (duration_hours: 96) by enabling multiple discharge cycles. Storage does not resolve the underlying conductor thermal limits on the MIS-F05 feeders and should be treated as a complement to reconductoring, not a substitute.

METHODOLOGY

Peak demand was estimated using a deterministic load model that applies electrification penetration rates (heat pump and EV shares) to the baseline winter peak of 3,855.4 MW (baseline_winter_peak_mw: 3855.4), with heat pump loads adjusted using COP curves that reflect degraded performance at temperatures approaching -32.5 degrees Celsius (peak_temp_c: -32.5). Per-feeder utilization values are modeled by distributing system-level load growth across individual feeder attributes including rated MVA, age, and current penetration levels; these are model outputs and should be validated against SCADA and metering records before use in capital planning. The network topology data (substations: 22, feeders: 176) is flagged as non-synthesized (is_synthesized: False), meaning it reflects actual Alectra asset counts rather than generated proxies.

SAFETY FLAGS

During a 96-hour cold event at -30.0 degrees Celsius, customers on the 88 at-risk feeders who depend on electrically powered medical devices (home oxygen concentrators, dialysis equipment, powered wheelchairs) face life-safety risk if thermal overload causes sustained outages. Alectra should cross-reference its medical baseline registry against the customer lists for at-risk feeders before the next heating season.

The 4,092 customers directly served by the three MIS-F05 feeders already operating at 117.3% utilization face the highest probability of outage onset at hour 43 of a sustained cold event, when temperatures are projected to reach -32.5 degrees Celsius. Vulnerable population outreach (elderly residents, low-income households without backup heat) should be coordinated with the City of Mississauga emergency management office.

A 96-hour outage duration during a -30.0 degree Celsius event exceeds the survivability window for uninsulated or poorly insulated dwellings without supplemental heat. If load shedding is required to protect system integrity, Alectra and municipal partners should pre-identify warming centres with capacity for the affected customer population.

LIMITATIONS

Only the Aggressive 2050 scenario (90% heat pump, 95% EV) was provided. Without a moderate or base-case electrification scenario, it is not possible to identify the penetration threshold at which the system transitions from passing to failing, which is the most actionable planning input for phased infrastructure investment.

Per-feeder utilization figures are outputs of a deterministic model and have not been validated against live SCADA measurements or historical peak loading records. Actual feeder headroom may differ materially, particularly for feeders with recent unrecorded load transfers or temporary switching configurations.

The analysis does not account for distributed energy resources (rooftop solar, battery storage, or vehicle-to-grid capability) that may be present on at-risk feeders by 2050 and could partially offset electrification-driven load growth. Omitting these resources makes the results conservative but potentially overstates required capital investment.

EXECUTIVE SUMMARY

Executive Summary: Mississauga Grid Faces Capacity Deficit Under Full Electrification

Headline Finding

The Aggressive 2050 scenario (90% heat pump, 95% EV penetration) produces a system-wide capacity deficit of 5116.6 MW (headroom_mw: -171.7), meaning Alectra Utilities' Mississauga network cannot reliably serve projected peak demand during a sustained cold event without intervention.

Scenario Evidence

Only one scenario was evaluated. Under the Aggressive 2050 case, modeled peak demand reaches 5,116.6 MW (peak_mw: 5116.6) against a baseline winter peak of 3,855.4 MW (baseline_winter_peak_mw: 3855.4), a increase of 1,261.2 MW or roughly 33%. The system stress peak occurs at hour 43 of the 96-hour event (peak_hour: 43) when the temperature drops to -32.5 degrees Celsius (peak_temp_c: -32.5), which is 2.5 degrees below the design event floor of -30.0 degrees Celsius. The scenario verdict is FAIL (verdict: FAIL) and the headroom percentage is -3.5% (headroom_pct: -3.5).

At-Risk Assets and Customers

Of 176 total feeders (feeders: 176), 88 are flagged as at risk (at_risk_feeders: 88), representing exactly half the distribution network. The three most stressed feeders, MIS-01-F05, MIS-02-F05, and MIS-03-F05, are each operating at 117.3% utilization (utilization_pct: 117.3) against a rated capacity of 30.66 MVA (rated_mva: 30.66). All three are 30 years old (age_years: 30) and have received no capacity upgrades (had_upgrade: false). Each feeder serves 1,364 customers (customers: 1364), putting a combined 4,092 customers at direct thermal overload risk on these three circuits alone. Across all 88 at-risk feeders, a proportionally larger share of the total 192,544 customers (customers: 192544) faces potential outage exposure during the modeled event.

Caveats and Limitations

This analysis models a single scenario at one extreme end of the electrification spectrum. No moderate or conservative electrification pathway was provided for comparison, so the range of plausible outcomes is not fully characterized here. Per-feeder load allocation is modeled deterministically and does not capture real-time switching, load transfer between feeders, or operator intervention. The cold event used (-30.0 degrees Celsius sustained for 96 hours) is a severe but not unprecedented design stress for southern Ontario; actual event probability and return period data are not included in the input data provided.

Recommended Next Action

Alectra should immediately commission a targeted capacity assessment of the 88 at-risk feeders, prioritizing the three MIS-F05 circuits given their current overload status independent of any future electrification growth. In parallel, a demand response enrollment program targeting EV charging deferral and heat pump cycling should be scoped for the 2025 to 2026 program year, as this is the fastest available tool to reduce peak exposure before capital upgrades can be completed.

DATA PROVENANCE

● CITY DEMOGRAPHICS

● NWS ALERTS

● SUBSTATION LOCATIONS

● UTILITY LOAD BASELINE

● COLD EVENT ARCHIVE

● FEEDER TOPOLOGY

● SCENARIO APPLICATION

● LOAD SIMULATION

● RISK RATIONALE

● SUMMARY

● LIVE API

● FROZEN SNAPSHOT

● MODELED

● LLM NARRATIVE

SOURCES

● FROZEN	Synthetic, derived from 2014 Polar Vortex envelope
● FROZEN	Alectra Utilities 2023 Annual Report
● FROZEN	IESO Yearly Reliability Outlook 2024
● FROZEN	StatsCan Census 2021, Mississauga CSD
● FROZEN	StatsCan 25-10-0015 (2026): 43,721,776 MWh annual generation in ON
● LIVE	ArcGIS Living Atlas NWS Watches & Warnings (live): no active alerts for Mississauga
● LIVE	ArcGIS GeoEnrichment (Esri Canada): Mississauga, ON (CY): 192,532 households, population 768,433

